

Habibur Rahaman

Ph.D. Candidate, Electrical and Computer Engineering, University of Florida

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EDUCATION

University of Florida

Ph.D., Electrical and Computer Engineering (GPA: 3.89)

Aug. 2022 – Present

Gainesville, FL, USA

Indian Institute of Technology, Kharagpur

M.S., Computer Science and Engineering (GPA: 3.78)

July 2019 – Jan. 2022

Kharagpur, India

RCC Institute of Information Technology

B.Tech., Computer Science and Engineering (GPA: 3.42)

July 2015 – May 2019

Kolkata, India

EXPERIENCE

University of Florida

Graduate Research Assistant, ECE Department (Advisor: Prof. Swarup Bhunia)

Aug. 2022 – Present

Gainesville, FL

- **ML Security Research:** investigating adversarial threats against neural networks and large language models, including white-box and black-box hyperparameter stealing, bit-flip attacks, subpopulation attacks, gradient sabotage, and steganographic weight embedding across PyTorch, ONNX, and TensorRT pipelines targeting top-tier security venues.
- **Runtime Attack Detection in AI Accelerators:** designed an on-chip framework based on AI Performance Counters and a lightweight machine learning analysis engine for runtime detection of adversarial attacks against deep neural network accelerators.

Intel Corporation (Remote)

Research Security Intern

Aug. 2023 – Dec. 2023

Santa Clara, CA

- Worked on the development and validation of countermeasures on Intel's Structured eASIC technology, and on the test vehicles used to validate the countermeasures of the Intel eASIC project.

Indian Institute of Technology, BHU (Varanasi)

Research Intern

Jan. 2022 – June 2022

Varanasi, India

- Developed an Emergency Evacuation System for shortest-safe path navigation in dynamic fire regions using IoT-enabled wireless sensor networks.

Intel Corporation

Research Intern, HPG Post-Validation DFT Group

June 2021 – Apr. 2022

Bangalore, India

- Built a machine learning model that predicts and fixes the nature of errors (syntax errors and module errors) occurring while compiling Verilog source files. The neural network localizes the fault and corrects the code without invoking the Synopsys VCS tool, after which the corrected source is passed to VCS for downstream processing.

Indian Institute of Technology, Kharagpur

Junior Project Officer (Sponsored by Synopsys Inc., USA)

June 2019 – Dec. 2019

Kharagpur, India

- Developed a machine learning-based testing technique for digital circuits as part of an EDA tool research initiative.

Indian Institute of Technology, Kharagpur

Research Intern

May 2018 – July 2018

Kharagpur, India

- Developed a distributed digital signature and security application using Blockchain technology.

SKILLS

Programming	C, C++, Python, Verilog, MATLAB
Deep Learning	PyTorch, TensorFlow, scikit-learn, ONNX, TensorRT, bitsandbytes, Hugging Face Transformers
EDA Tools	Synopsys Design Compiler, Synopsys VCS, Cadence Genus, Cadence JasperGold, Cadence Virtuoso, Cadence LEC, Xilinx Vivado, Design Navigator (Intel)

SELECTED RESEARCH PROJECTS

SAMURAI – Runtime Attack Detection in AI Accelerators. A framework for runtime detection of adversarial (evasion) attacks against AI accelerators. SAMURAI integrates an array of on-chip AI Performance Counters (APCs) with a lightweight on-chip ML analysis engine, TANTO (Trained Anomaly Inspection through Trace Observation), to monitor low-level hardware events during AI inference and flag security threats. Evaluated on image and audio (raw and Mel-spectrogram) inputs, achieving 98% detection accuracy with <3% energy, <4% area, and <10% inference latency overhead.

BARE-AI – Bit-Flip Attack Resilience in AI Hardware. A real-time, low-overhead framework that detects, localizes, and mitigates Bit-Flip Attacks (BFAs) on DNN inference workloads. BARE-AI uses APCs to capture fine-grained runtime features that are concurrently analyzed by PULSE (Predictive Unit for Layer Security Evaluation), a lightweight ensemble of ML classifiers. An Activation Shift Index (ASI) provides layer-level fault localization, and a z-score-based statistical repair restores corrupted parameters. Achieves up to 98% detection accuracy across CNNs, Vision Transformers, and Large Language Models against random, targeted, and adaptive BFAs, with <3% energy, <4% area, and ~10% latency overhead.

WINGs – Automatic Weight Generator for Secure and Storage-Efficient DNNs. A framework that dynamically generates fully connected layer weights and compresses convolutional layer weights at inference time, eliminating full-weight matrix storage. WINGs uses Principal Component Analysis (PCA) for dimensionality reduction and lightweight Support Vector Regression (SVR) models to predict weights, with sensitivity-aware compression on low-sensitivity CNN layers. Delivers 53× compression on FC layers, 28× on AlexNet/MNIST, and 18× on AlexNet/CIFAR-10 with 1–2% accuracy loss. Sensitivity-aware design also amplifies and exposes the impact of bit-flip attacks on compressed weights.

DASH – Meta-Attack Framework for Perceptually Aligned Adversarial Examples. A differentiable, multi-stage meta-attack framework that composes existing ℓ_p -bounded attacks via learned adaptive weights to produce effective and perceptually aligned adversarial examples. A novel meta-loss jointly minimizes misclassification loss and perceptual distortion (SSIM, LPIPS, FID). Evaluated on adversarially trained models on CIFAR-10, CIFAR-100, and ImageNet, DASH outperforms state-of-the-art perceptual attacks (e.g., AdvAD), with ~20.63% higher attack success rate and substantial improvements in SSIM, LPIPS, and FID.

E³-CODE – Embedded Error-Correcting Code for Error-Resilient Neural Networks. An ECC framework that embeds multi-bit parity within the entire weight representation rather than overwriting only LSBs. Weight and parity assignments are jointly optimized through a Mixed-Integer Linear Programming (MILP) formulation, and a hybrid scheme combines embedded ECC with conventional ECC. Evaluated on ResNet, MobileNetV2, and EfficientNet-B0 on ImageNet and CIFAR-10, E³-CODE preserves software-level accuracy under varying Bit Error Rates, extending edge-system lifetime by 4.3× with no memory overhead and 10× with <2% memory overhead.

PUBLICATIONS

Journal Articles – Published or Accepted

- **H. Rahaman**, A. Chatterjee, and S. Bhunia. “SAMURAI: Runtime Attack Detection in AI Accelerators Using AI Performance Counters.” *IEEE Transactions on Circuits and Systems for Artificial Intelligence*, 2026. [IEEE Xplore](#)
- A. Chatterjee, **H. Rahaman**, and S. Bhunia. “ASTRA: Automated Insertion of Distributed Entropy Sources for Robust Authentication.” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 34, no. 2, pp. 634–647, 2025.
- **H. Rahaman**, A. Chatterjee, and S. Bhunia. “Fortifying AI Systems: Emerging Threats and Security Countermeasures.” *Springer Nature Computer Science*, vol. 7, no. 227, 2025. doi:[10.1007/s42979-025-04658-y](https://doi.org/10.1007/s42979-025-04658-y).

Journal Articles – Under Review or Revision

- **H. Rahaman**, A. Chatterjee, and S. Bhunia. “WINGS: Automatic Weight Generator for Secure and Storage-Efficient Deep Learning Models.” *IEEE Transactions on Computers*, 2025 (Major Revision, Submitted). Preprint: [arXiv:2507.06380](https://arxiv.org/abs/2507.06380).
- **H. Rahaman**, S. Bhattacharya, S. Das, K. Basu, and S. Bhunia. “BAREAI: Bit-Flip Attack Resilience in AI Hardware through Built-in Performance Monitors.” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* (Submitted).
- S. Paria, Z. Han, A. Dasgupta, **H. Rahaman**, D. Halder, R. Almawzan, D. Xing, K. Amberiadis, S. Ray, J. Rajendran, A. Srivastava, and S. Bhunia. “Benchmarking of Secure Hardware: Evaluation of Logic Locking, State Space Transformation, Redaction, and NoC Routing Table Configuration.” *ACM Transactions on Design Automation of Electronic Systems*, 2026 (Minor Revision).

Conference Papers – Accepted

- A.A.N. Nafi, **H. Rahaman**, Z. Haider, T. Mahfuz, F. Suya, S. Bhunia, and P. Chakraborty. “DASH: A Meta-Attack Framework for Synthesizing Effective and Stealthy Adversarial Examples.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. Preprint: [arXiv:2508.13309](https://arxiv.org/abs/2508.13309).
- **H. Rahaman** et al. “ENTROMIX: Clinically-Aware Self-Paced Generative Augmentation for Imbalanced Medical Image Classification.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- N. Kamineni, **H. Rahaman**, S. Bhunia, and R. Ewetz. “E³-CODE: Embedded and Efficient Error-Correcting Code for Error-Resilient Neural Networks.” *63rd Design Automation Conference (DAC)*, July 26–29, 2026, Long Beach, CA.
- **H. Rahaman**, S. Paria, A. Chatterjee, and S. Bhunia. “Evolving Landscape of Attacks on AI Hardware and Robust Defenses.” *27th International Symposium on Quality Electronic Design (ISQED)*, 2026.
- A. Chatterjee, S. Paria, **H. Rahaman**, B. Chatterjee, and S. Bhunia. “CRISP: A Unified, Configurable, Resilient Infrastructure for Security Primitives with $0.071 \times 10^6 F^2/\text{bit}$ Feature Size.” *27th International Symposium on Quality Electronic Design (ISQED)*, 2026.
- A. Chatterjee, **H. Rahaman**, and S. Bhunia. “MARVEL-PUF: A Robust Multi-Bit Memory PUF for FPGA-based Embedded Systems Security.” *International Conference on Codesign and System Synthesis (CODES+ISSS)*, December 2025. doi:[10.1145/3742873.3755988](https://doi.org/10.1145/3742873.3755988).
- **H. Rahaman**, A. Chatterjee, and S. Bhunia. “SAMURAI: A Framework for Safeguarding Against Malicious Usage and Resilience of AI.” *IEEE 33rd Asian Test Symposium (ATS)*, pp. 1–6, 2024. doi:[10.1109/ATS64447.2024.10915409](https://doi.org/10.1109/ATS64447.2024.10915409). [Best Paper Nominee]
- **H. Rahaman**, A. Chatterjee, and S. Bhunia. “Secure AI Systems: Emerging Threats and Defense Mechanisms.” *IEEE 33rd Asian Test Symposium (ATS)*, pp. 1–6, 2024. doi:[10.1109/ATS64447.2024.10915428](https://doi.org/10.1109/ATS64447.2024.10915428). [IEEE Xplore](https://ieeexplore.ieee.org/document/10915428)

PATENTS

- **H. Rahaman**, S. Bhunia, B. Chatterjee, and A. Chatterjee. “Pre-Computation-Based Implementation of Neural Networks.” US Patent Application US20260093966A1, published April 2, 2026.
- **H. Rahaman**, S. Bhunia, A. Dasgupta, A. Chatterjee, and S. Paria. “Reconfigurable Security Fabric for Securing Digital Circuit Designs.” US Patent Application US20250384197A1, published December 18, 2025.
- “MBM-PUF: A Multi-Bit Memory-based Physically Unclonable Function.” (Patent Pending)
- “HAL: Hardware-Based Licensing of AI Chips.” (Patent Pending)

AWARDS AND HONORS

- **Best Demo Award**, UF Innovate, Warren B. Nelms Annual IoT Conference 2025, for the SAMURAI tool demonstration.
- **First Prize, Research Forum Presentation**, AIHOT 2025.
- **Best Paper Nominee**, IEEE 33rd Asian Test Symposium (ATS), 2024, for the SAMURAI framework paper.

TEACHING AND MENTORSHIP

- **Teaching Assistant**, “Special Topics in Freshman Engineering: Adventures in ECE Design,” University of Florida, Fall 2023.
- **Teaching Assistant**, “VLSI Circuits and Technology 2” (graduate course), University of Florida, Spring 2024.
- **Student Mentor**, undergraduate research mentee, University of Florida, Summer 2024.
- **Teaching Assistant**, NPTEL India: Compiler Design, Operating Systems, and Digital Circuits.

PROFESSIONAL SERVICE

Reviewer: International Symposium on Devices, Circuits and Systems (ISDCS); Conference on Secure IoT, Assured and Trusted Computing; IEEE Custom Integrated Circuits Conference; VLSI Design Conference.

REFERENCES

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